

The Root Space Decomposition

1.1 Preliminary Results

Proposition 1.1.1

Suppose that L is a complex semisimple Lie algebra containing an abelian subalgebra H consisting of semisimple elements. For each $\alpha \in H^*$, let

$$L_\alpha := \{x \in L : [h, x] = \alpha(h)x, \quad \forall h \in H\}$$

Let Φ denote the set of non-zero $\alpha \in L^*$ for which L_α is non-zero. We can write the decomposition of L into weight spaces for H as

$$L = L_0 \oplus \bigoplus_{\alpha \in \Phi} L_\alpha. \quad (\star)$$

Since L is finite-dimensional, this implies that Φ is finite.

Lemma 1.1.2

Suppose $\alpha, \beta \in H^*$, then

1. $[L_\alpha, L_\beta] = L_{\alpha+\beta}$
2. If $\alpha + \beta \neq 0$, then $\kappa(L_\alpha, L_\beta) = 0$.
3. The restriction of κ to L_0 is non-degenerate.

Proof. 1. Take $x \in L_\alpha, y \in L_\beta$, then

$$\begin{aligned} [h, [x, y]] &= -[y, [h, x]] - [x, [y, h]] \\ &= -[y, \alpha x] - [x, -\beta y] \\ &= (\alpha + \beta)[x, y] \end{aligned}$$

2. If $\alpha + \beta \neq 0$, then $\alpha + \beta \neq 0$. Take $x \in L_\alpha, y \in L_\beta$

$$\begin{aligned}
 \alpha(h) \kappa(x, y) &= \kappa(\alpha(h)x, y) \\
 &= \kappa([h, x], y) \\
 &= -\kappa([x, h], y) \\
 &= -\kappa(x, [h, y]) \\
 &= -\kappa(x, \beta(h)y) \\
 &= -\beta(h) \kappa(x, y)
 \end{aligned}$$

Thus

$$(\alpha(h) + \beta(h)) \kappa(x, y) = 0$$

Then it follows that $\kappa(x, y) = 0$ since $\alpha(h) + \beta(h) \neq 0$

3. Suppose that $z \in L_0, \kappa(z, x_0) = 0$ for all $x_0 \in L_0$. From 2., for each $\alpha \neq 0$, we have $\kappa(z, x_\alpha) = 0$ for all $x_\alpha \in L_\alpha$. Then for each element of x , we write

$$x = x_0 + \sum_{\alpha \in \Phi} x_\alpha$$

there is

$$\kappa(z, x) = \kappa(z, x_0) + \sum_{\alpha \in \Phi} \kappa(z, x_\alpha) = 0$$

Since L is semisimple, κ is non-degenerate (Cartan second Criterion). It follows that $z = 0$.

□

Exercise 1.1.3

Show that if $x \in L_\alpha$, where $\alpha \neq 0$, then ad_x is nilpotent.

Proof. Suppose that $\Phi = \{\alpha_0 := \alpha, \alpha_1, \dots, \alpha_k\}$, then for each $j = 0, 1, \dots, k$, since $\alpha_j \neq 0$, there exists $m_j \in \mathbb{Z}_{>0}$ such that $\alpha_j + m_j \alpha$ is different from any element of $\Phi \cup \{0\}$. Then

$$(\text{ad}_x)^{m_j}(x_{\alpha_j}) = 0, \quad \forall j = 0, 1, \dots, k, \quad \forall x_{\alpha_j} \in L_{\alpha_j}$$

Furthermore,

$$(\text{ad}_x)^{m_0+1}(x_0) = (\text{ad}_x)^{m_0}([x, x_0]) = 0, \quad \forall x_0 \in L_0$$

since $[x, x_0] \in L_\alpha$. Set $N = \max\{m_0 + 1, m_1, \dots, m_k\}$. Then

$$(\text{ad}_x)^N(L) = (\text{ad}_x)^N(L_0 + \bigoplus_{\alpha_j \in \Phi} L_{\alpha_j}) = 0$$

ad_x is nilpotent. □

Remark.

If H is small, then the decomposition (★) is likely to be rather coarse, with few non-zero weight spaces other than L_0 . Furthermore, if H is properly contained in L_0 , then we get little information about how the elements in L_0 that are not in H act on L . This is illustrated by the following exercise.

Exercise 1.1.4

Let $L = \mathfrak{sl}(n, \mathbb{C})$, where $n \geq 2$, and let $H = \text{Span}\{h\}$, where $h = e_{11} - e_{22}$. Find $L_0 = C_L(H)$, and determine the direct sum decomposition (★) with respect to H .

Solution. Take $x \in \mathfrak{sl}(n, \mathbb{C})$, suppose $x = \sum_{i \neq j, \text{ and } i, j \neq 1, 2} x^{ij} x_{ij} + x^{12} e_{12} + x^{21} e_{21} + y^k \sum_{k=2}^n (e_{11} - e_{kk})$. For each $1 \leq i \neq j \leq n$ If $i, j \neq 1, 2$, then

$$[e_{11}, e_{ij}] = e_{11}e_{ij} - e_{ij}e_{11} = 0$$

$$[e_{22}, e_{ij}] = e_{22}e_{ij} - e_{ij}e_{22} = 0$$

$$[h, x^{12}e_{12}] = x^{12}[h, e_{12}] = x^{12}(e_{11}e_{12} - e_{12}e_{11} - e_{22}e_{12} + e_{12}e_{22}) = 2x^{12}e_{12}$$

$$[h, x^{21}e_{21}] = x^{21}[h, e_{21}] = x^{21}(e_{11}e_{21} - e_{21}e_{11} - e_{22}e_{21} + e_{21}e_{22}) = -2x^{21}e_{21}$$

$$[h, e_{11} - e_{kk}] = [e_{22}, e_{kk}] = 0$$

Then $x \in L_0 \iff x^{12} = x^{21} = 0$.

$$L_0 = \text{span}(\{e_{ij}, i \neq j, \text{ and } i, j \neq 1, 2\} \cup \{e_{12} - e_{21}\} \cup \{e_{11} - e_{kk} : k = 2, 3, \dots, n\})$$

◇

1.2 Cartan Subalgebras

Definition 1.2.1

A Lie subalgebra H of a Lie algebra L is said to be a **Cartan subalgebra** (or CSA) if H is abelian and every element $h \in H$ is semisimple, and moreover H is maximal with these properties.

Lemma 1.2.2

Let H be a Cartan subalgebra of L . Suppose that $h \in H$ is such that the dimension of $C_L(h)$ is minimal. Then every $s \in H$ is central in $C_L(h)$, and so $C_L(h) \subseteq C_L(s)$. Hence $C_L(h) = C_L(H)$.

Theorem 1.2.3

If H is a Cartan subalgebra of L and $h \in H$ is such that $C_L(h) = C_L(H)$, then $C_L(h) = H$. Hence H is self-centralising.

1.3 Definition of the Root Space Decomposition

Definition 1.3.1

Let H be a Cartan subalgebra of our semisimple Lie algebra L . As $H = C_L(H)$, the direct sum decomposition of L into weight spaces for H considered in §10.1 may be written as

$$L = H \oplus \bigoplus_{\alpha \in \Phi} L_{\alpha},$$

where Φ is the set of $\alpha \in H^*$ such that $\alpha \neq 0$ and $L_{\alpha} \neq 0$. Since L is finite-dimensional, Φ is finite.

- If $\alpha \in \Phi$, then we say that α is a **root** of L and L_{α} is the associated **root space**.
- The direct sum decomposition above is the **root space decomposition**.

Remark.

It should be noted that the roots and root spaces depend on the choice of Cartan subalgebra H .

Lemma 1.3.2

Let $x, y : V \rightarrow V$ be linear maps from a complex vector space V to itself. Suppose that x and y both commute with $[x, y]$. Then $[x, y]$ is a nilpotent map.

Proof. Let $S = \text{span}\{x, y\}$, then

$$S' = \text{span}\{[x, y]\}$$

commutes with all element of S , that is $[S', S] = 0$, it follows that S is nilpotent, thus it is solvable. From Lie's theorem, there is a basis of V in which x, y is represented upper triangular. Let $L_i = \text{span}\{e_1, \dots, e_i\}$, then L_i is x -invariant and y -invariant, $i = 1, \dots, n$. There is $x^{ii}, y_{ii}, i = 1, \dots, n$ such that

$$x(e_i) + L_{i-1} = x^{ii}e_i, \quad y(e_i) + L_{i-1} = y^{ii}e_i.$$

Then

$$[x, y](e_i) + L_{i-1} = x^{ii}y^{ii}e_i - y^{ii}x^{ii}e_i + L_{i-1} = 0 + L_{i-1}$$

Thus $[x, y](e_i) \in L_{i-1}$. Thus $[x, y]$ is nilpotent. $\square$

Lemma 1.3.3

Suppose that $\alpha \in \Phi$ and that x is a non-zero element in L_α . Then $-\alpha$ is a root and there exists $y \in L_{-\alpha}$ such that $\text{Span}\{x, y, [x, y]\}$ is a Lie subalgebra of L isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Proof. κ is non-degenerate, there exists $w = y_0 + \sum_{\beta \in \Phi} y_\beta$ such that $\kappa(x, w) \neq 0$. But for each $\beta \neq \alpha$, $\kappa(x, y_\beta) = 0$, there must be $-\alpha \in \Phi$, $\kappa(x, y_{-\alpha}) = \kappa(x, w) \neq 0$. We take $y = y_{-\alpha}$. Since $\alpha \neq 0$, there exists $t \in H$ such that $\alpha(t) \neq 0$. Then

$$\kappa(t, [x, y]) = \kappa([t, x], y) = \alpha(t) \kappa(x, y) \neq 0$$

thus $[x, y] \neq 0$.

Let $S := \text{span}\{x, y, [x, y]\}$, $[x, y]$ lies in $L_0 = H$. As x and y are simultaneous eigenvectors for all elements of $\text{ad } H$, and so in particular for $\text{ad}[x, y]$, this shows that S is a Lie subalgebra of L . It remains to show that S is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

Let $h := [x, y] \in S$. We claim that $\alpha(h) \neq 0$. If not, then $[h, x] = \alpha(h)x = 0$;

similarly $[x, y] = -\alpha(h)x = 0$, so $\text{ad } h : L \rightarrow L$ commutes with $\text{ad } x$ and $\text{ad } y$, thus is nilpotent. The other hand, because H is a Cartan subalgebra, h is semisimple. But the only element of L that is both semisimple and nilpotent is 0, so $h = 0$, a contradiction.

Thus S is a 3-dimensional complex Lie algebra with $S' = S$. S is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$. $\square$

Exercise 1.3.4

Show that for each $\alpha \in \Phi$, S_α has a basis $\{e_\alpha, f_\alpha, h_\alpha\}$ such that

1. $e_\alpha \in L_\alpha, f_\alpha \in L_{-\alpha}, h_\alpha \in H$ and $\alpha(h_\alpha) = 2$.
2. The map $\theta : S_\alpha \rightarrow \mathfrak{sl}(2, \mathbb{C})$, defined by $\theta(e_\alpha) = e, \theta(f_\alpha) = f, \theta(h_\alpha) = h$ is a Lie algebra isomorphism.

Proof. Let $x \in L_\alpha$. Since κ is non-degenerate, there exists $w = y_0 + \sum_{\alpha \in \Phi} y_\alpha$ such that $\kappa(x, w) \neq 0$, then

$$\kappa(x, w) = \kappa\left(x, y_0 + \sum_{\alpha \in \Phi} y_\alpha\right) = \kappa(x, y_{-\alpha}) \neq 0$$

Denote $y = y_{-\alpha}$.

$[x, y] \in L_{\alpha+(-\alpha)} = L_0 = H$. Take $h = [x, y]$. $h(x) = \alpha(h)x$

$$\kappa(h(x), y) = \alpha(h)\kappa(x, y) \neq 0$$

Then $\alpha(h) \neq 0$. Take $e_\alpha = x, f_\alpha = \frac{2}{\alpha(h)}y, h_\alpha = [e_\alpha, f_\alpha]$. Then

$$h_\alpha(e_\alpha) = \frac{2}{\alpha(h)}h(x) = 2x = 2e_\alpha$$

Let $S_\alpha = \text{span}\{e_\alpha, f_\alpha, h_\alpha\}$. Then $S'_\alpha = S_\alpha$. S is isomorphic to $\mathfrak{sl}(2, \mathbb{C})$.

$$[e_\alpha, f_\alpha] = h_\alpha, \quad [h_\alpha, e_\alpha] = 2e_\alpha, \quad [h_\alpha, f_\alpha] = 2f_\alpha$$

$\square$

1.4 Root Strings and Eigenvalues

Lemma 1.4.1

Given $h \in H$, let θ_h denote the map $\theta_h \in H^*$ defined by

$$\theta_h(k) = \kappa(h, k), \quad \forall k \in H$$

Then it is an isomorphism since $\kappa|_H$ is non-degenerate. In particular, for $\alpha \in H^*$, there is a unique element $t_\alpha \in H$ such that

$$\kappa(t_\alpha, k) = \alpha(k), \quad \forall k \in H$$

Lemma 1.4.2

Let $\alpha \in \Phi$. If $x \in L_\alpha$, $y \in L_{-\alpha}$, then $[x, y] = \kappa(x, y)t_\alpha$. In particular $h_\alpha = [e_\alpha, f_\alpha] \in \text{span}\{t_\alpha\}$.

Proof. For $h \in H$, we have

$$\kappa(h, [x, y]) = \alpha(h)\kappa(x, y) = \kappa(t_\alpha, h)\kappa(x, y) = \kappa(h, \kappa(x, y)t_\alpha)$$

This shows that $[x, y] - \kappa(x, y)t_\alpha \in H^\perp = 0$. Thus $[x, y] = \kappa(x, y)t_\alpha$ □

Definition 1.4.3

Let α be a root. We may regard L as an $\mathfrak{sl}(\alpha)$ -module via restriction of the adjoint representation. Thus if $a \in \mathfrak{sl}(\alpha)$ and $y \in L$, then the action is given by

$$a \cdot y = (\text{ad } a)(y) = [a, y]$$

Remark.

- The $\mathfrak{sl}(\alpha)$ -submodules of L are precisely the vector subspaces M of L such that $[s, m] \in M$ for all $s \in \mathfrak{sl}(\alpha)$ and $m \in M$.

Lemma 1.4.4

If M is a $\mathfrak{sl}(\alpha)$ -submodule of L , then the eigenvalues of h_α acting on M are integers.

Proof. Since M is finite-dimensional, by Weyl's Theorem, M may be decomposed into a direct sum of irreducible $\mathfrak{sl}(\alpha)$ -modules; for irreducible $\mathfrak{sl}(2, \mathbb{C})$ -

modules, the eigenvalues of h acting on M are integers. The lemma follows from the isomorphism between $\mathfrak{sl}(\alpha)$ and $\mathfrak{sl}(2, \mathbb{C})$. $\square$

Definition 1.4.5

If $\beta \in \Phi$ or $\beta = 0$, let

$$M := \bigoplus_c L_{\beta+c\alpha}$$

where the sum is over all $c \in \mathbb{C}$ such that $\beta + c\alpha \in \Phi$. M is an $\mathfrak{sl}(\alpha)$ -submodule of L . This module is said to be the α -*root string through* $m\beta$.

Proof. For each $c_1, c_2 \in \mathbb{C}$. If $\beta \neq 0$, then $[L_{\beta+c_1\alpha}, L_{\beta+c_2\alpha}] \in L_{\beta+(\beta^{-1}\alpha+c_1+c_2)\alpha}$. If $\beta = 0$, then $[L_{\beta+c_1\alpha}, L_{\beta+c_2\alpha}] \in L_{\beta+(c_1+c_2)\alpha}$. $\square$

Proposition 1.4.6

Let $\alpha \in \Phi$. The root spaces $L_{\pm\alpha}$ are 1-dimensional. Moreover, the only multiples of α which lie in Φ are $\pm\alpha$.